Hostel Maintenance & Issue Ticketing System — UML Component Diagram (Layered Architecture)

Presentation Layer

«component»
Resident Portal
raise ticket · upload photo · view own tickets

«component»
Warden & Staff Console
intake dashboard · assign technician · update job · sign-off closure

«use»

Business Layer

ITicketSubmission
HTTPS / REST API

IIntakeDashboard
HTTPS / REST API

IDispatchControl
HTTPS / REST API

IStatusUpdate
HTTPS / REST API

ISignOffClosure
HTTPS / REST API

«component»
Ticket Intake & Triage Service
FR-001, FR-005

INewTicketFeed
internal API call

«component»
Staff Dispatch Service
FR-002

IWorkOrder
internal API call

«component»
Job Status Tracking Service
FR-003

IStatusEvent
internal event

«component»
SLA & Closure Service
FR-004, NFR-001

IDataAccess
repository API (ORM) · role-checked

Data Layer

«component»
Data Access & Media Storage Component
RBAC check · validation · AES-256 encryption

IStorage
SQL (JDBC) + object-store API

«component» «database»
Relational Database & Media Store
tickets · staff · work orders · photos

«use»

Legend

- Provided interface (ball)
- Required interface (socket)
- Assembly connector (solid line)
- «use» dependency between layers
- Layer package (strict adjacent-layer access)

Data flow: requests travel Presentation → Business → Data; results return the same path.